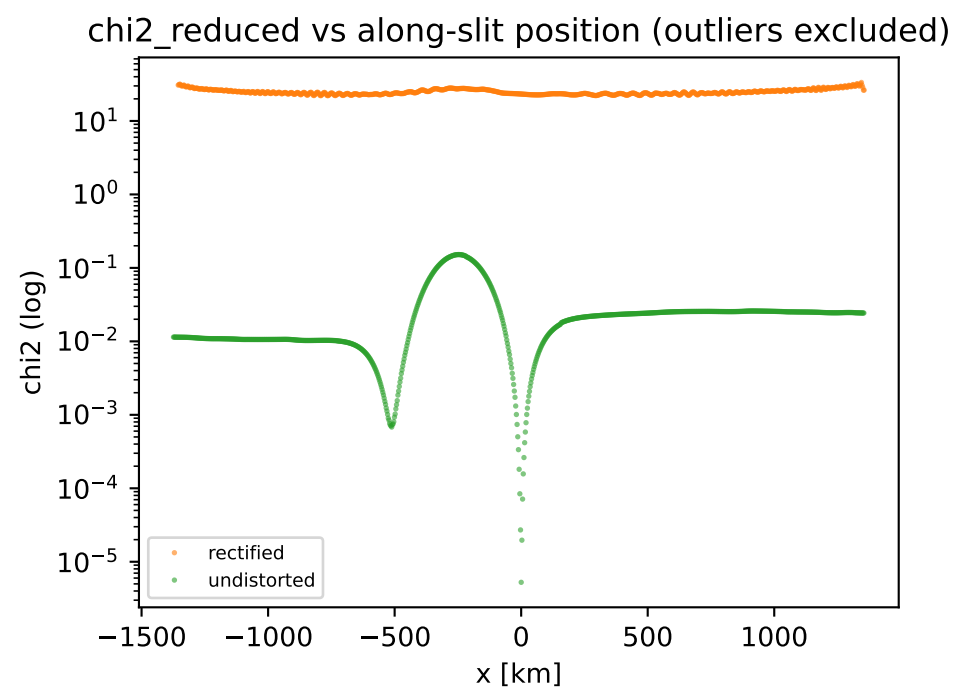
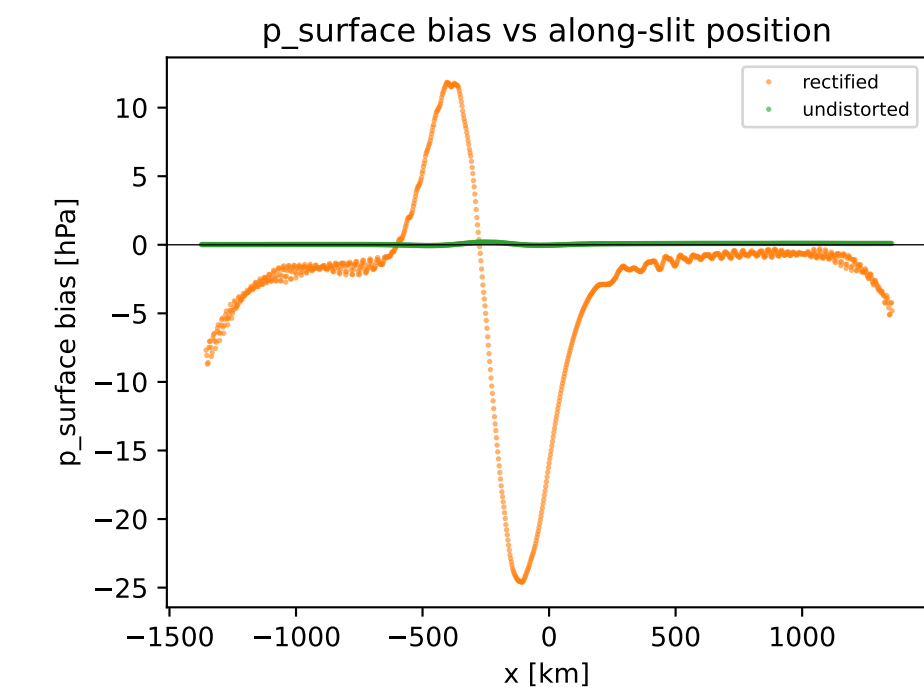
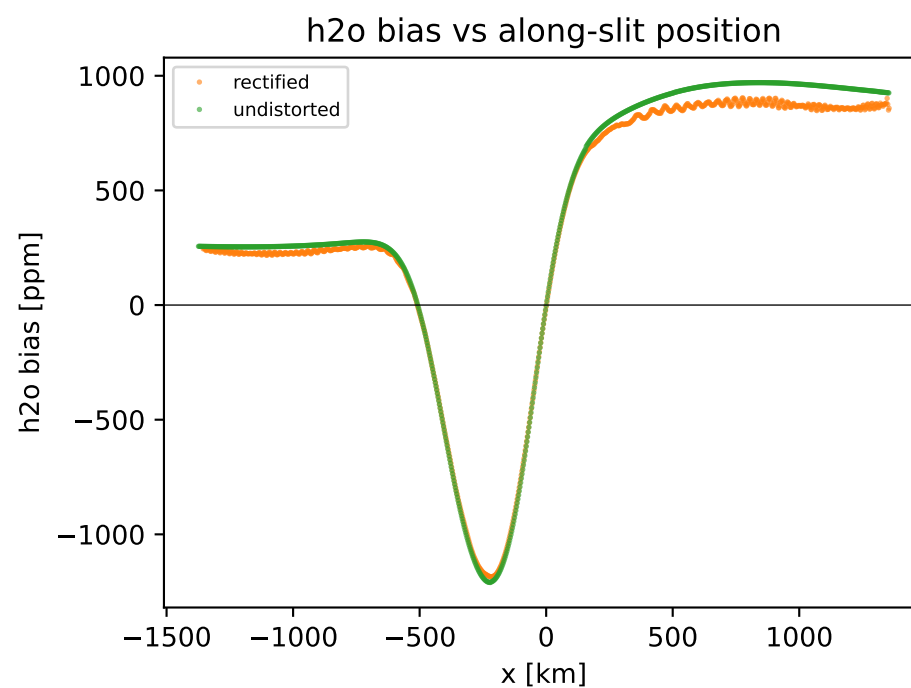
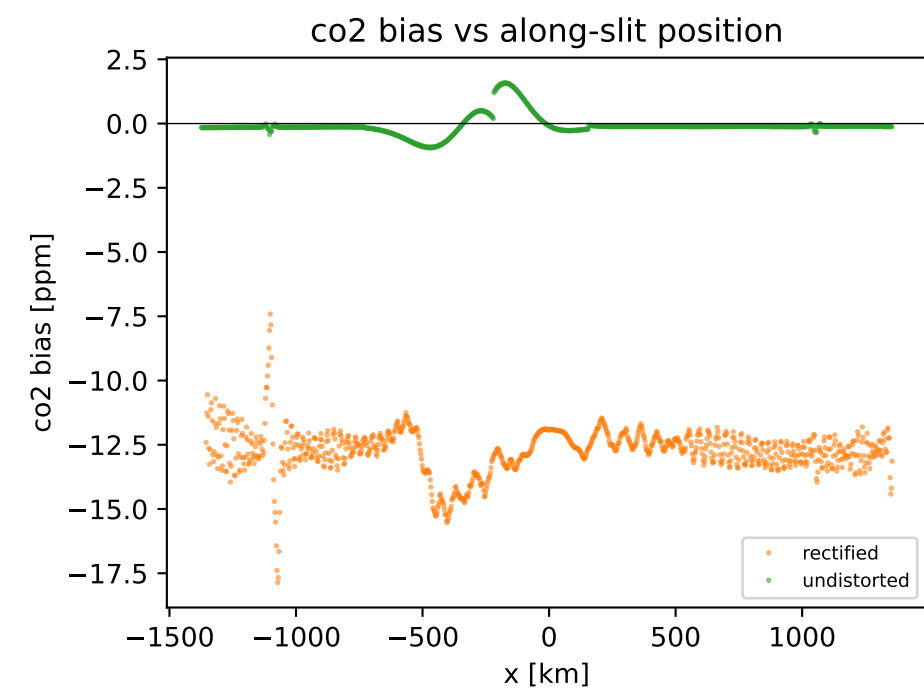
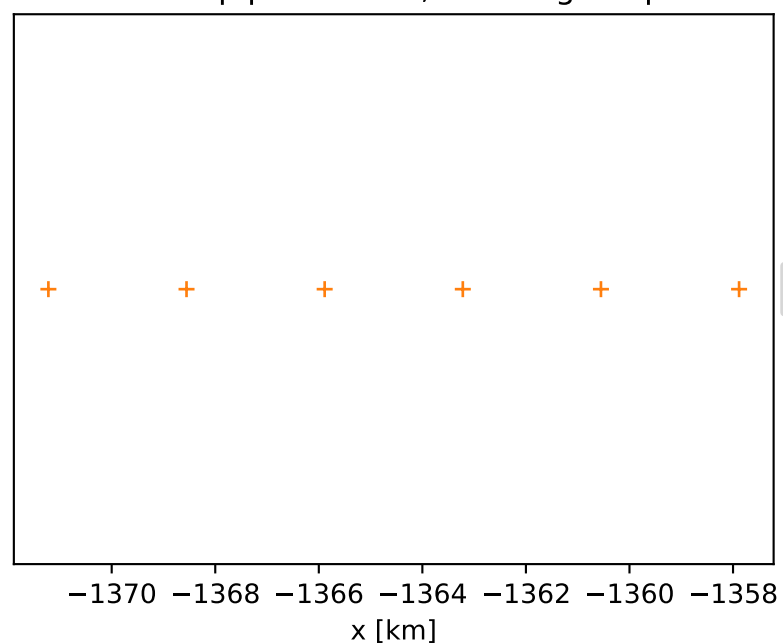


Joint FPA0+FPA2 -- realistic, no noise (rectified/undistorted)

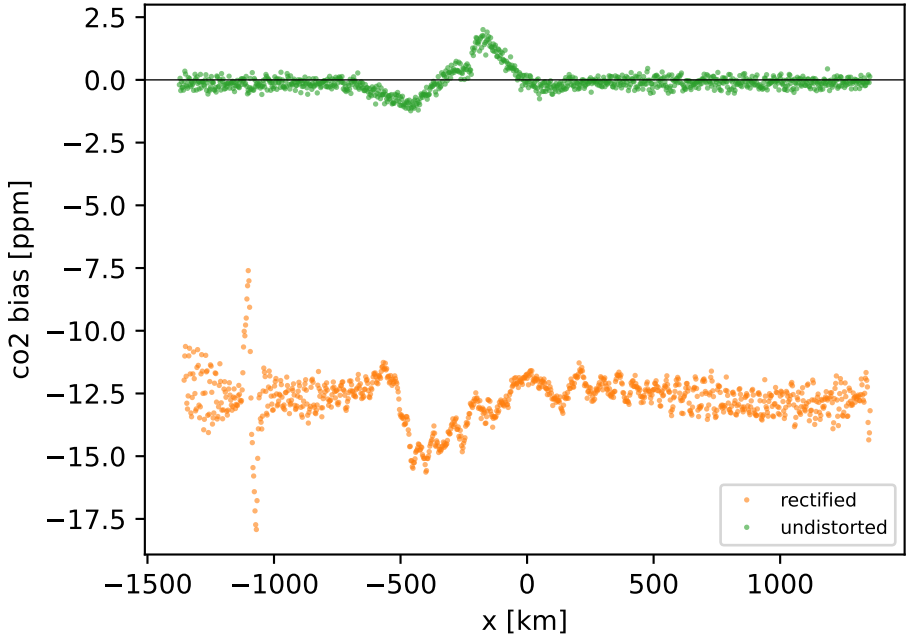


Where each pipeline fails, vs along-slit position

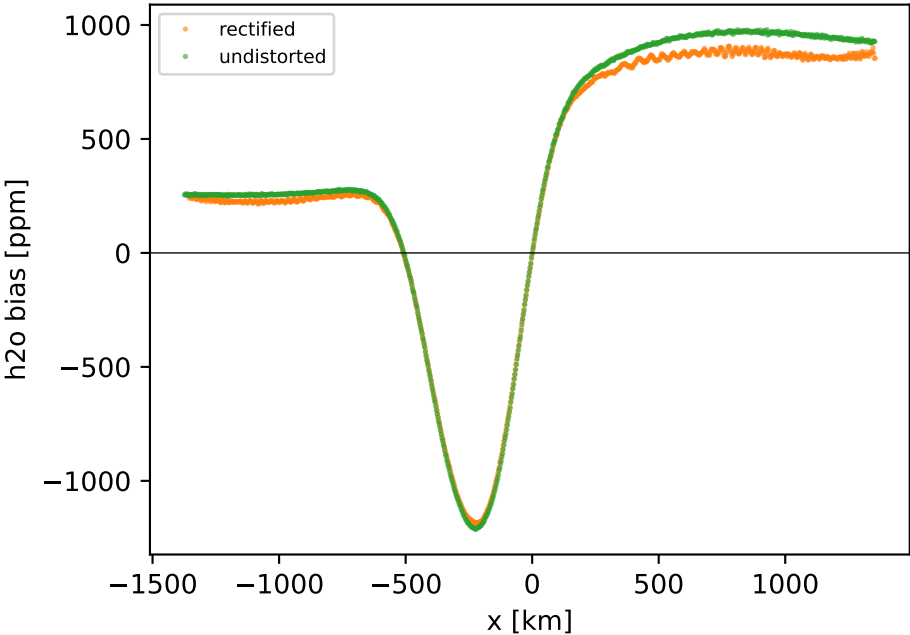


Joint FPA0+FPA2 -- realistic, noise (rectified/undistorted)

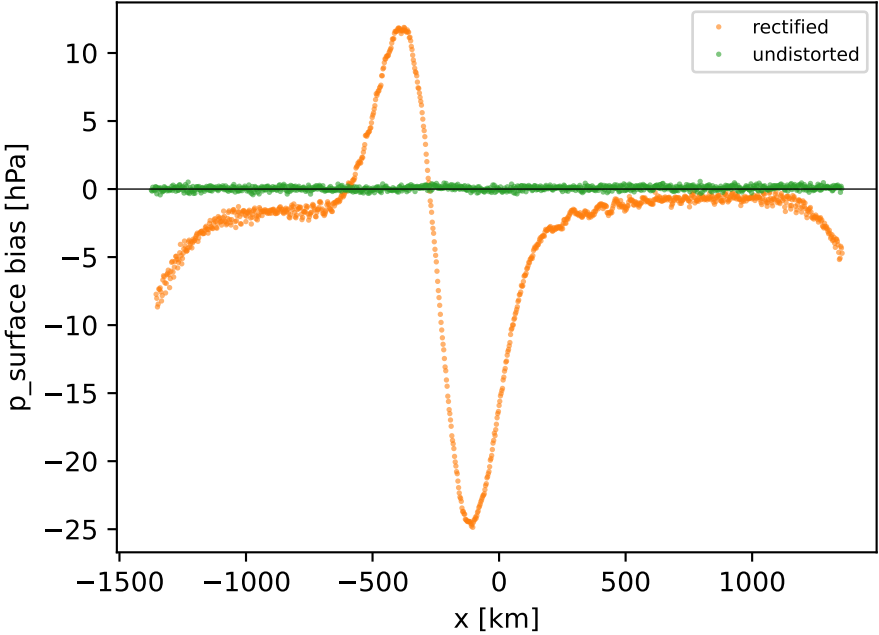
co2 bias vs along-slit position



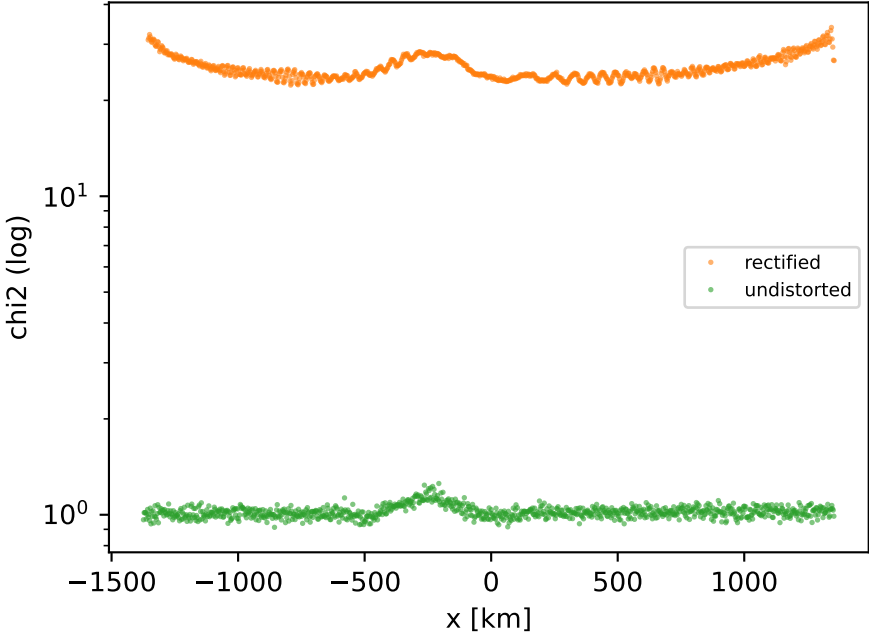
h2o bias vs along-slit position



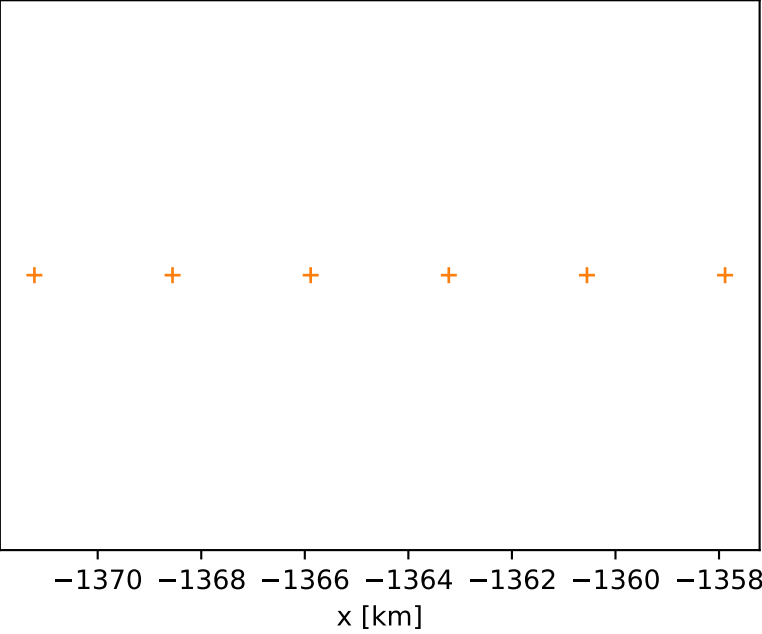
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

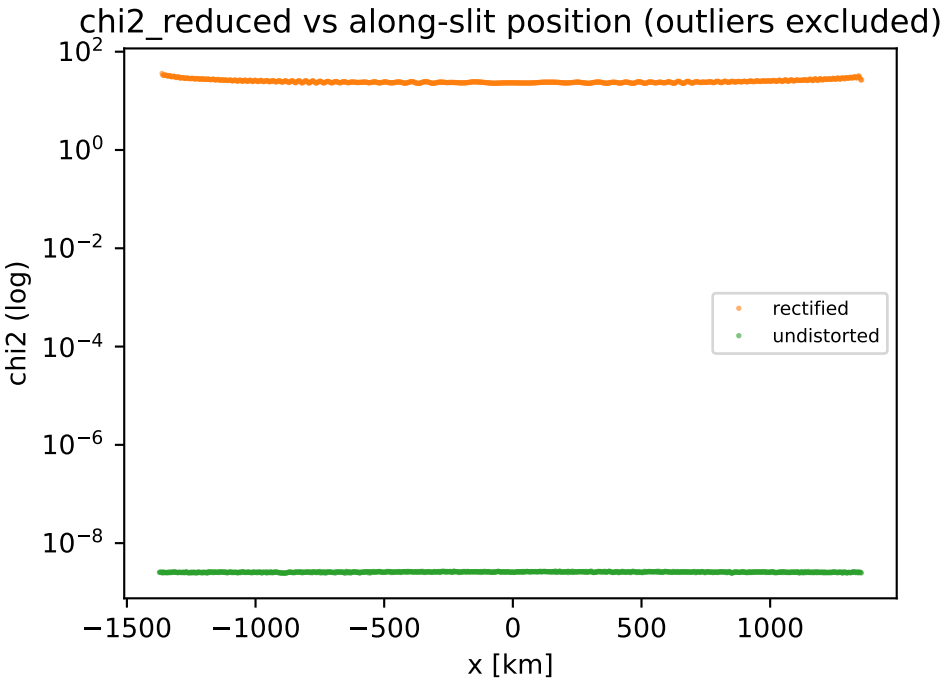
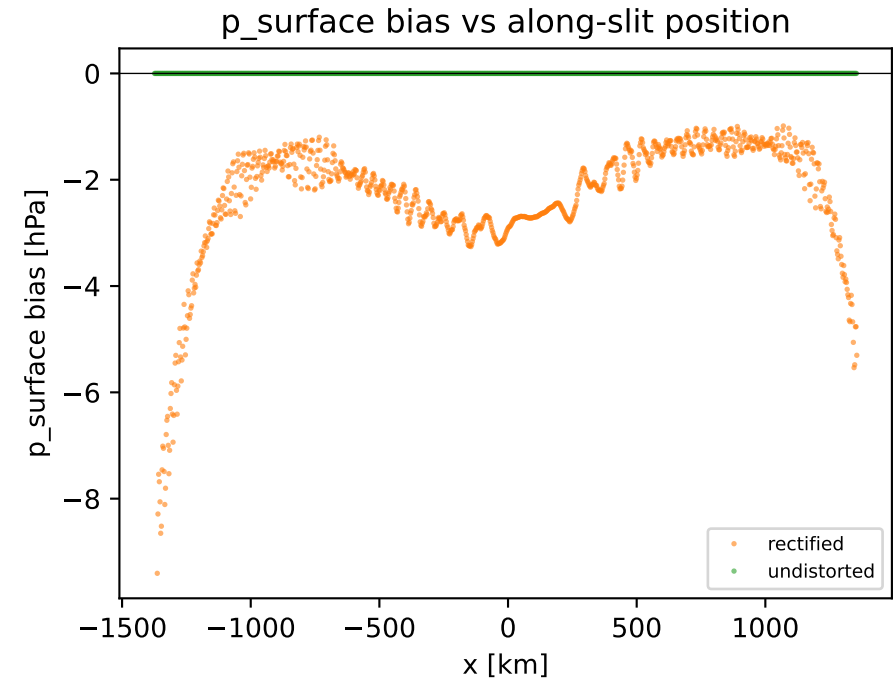
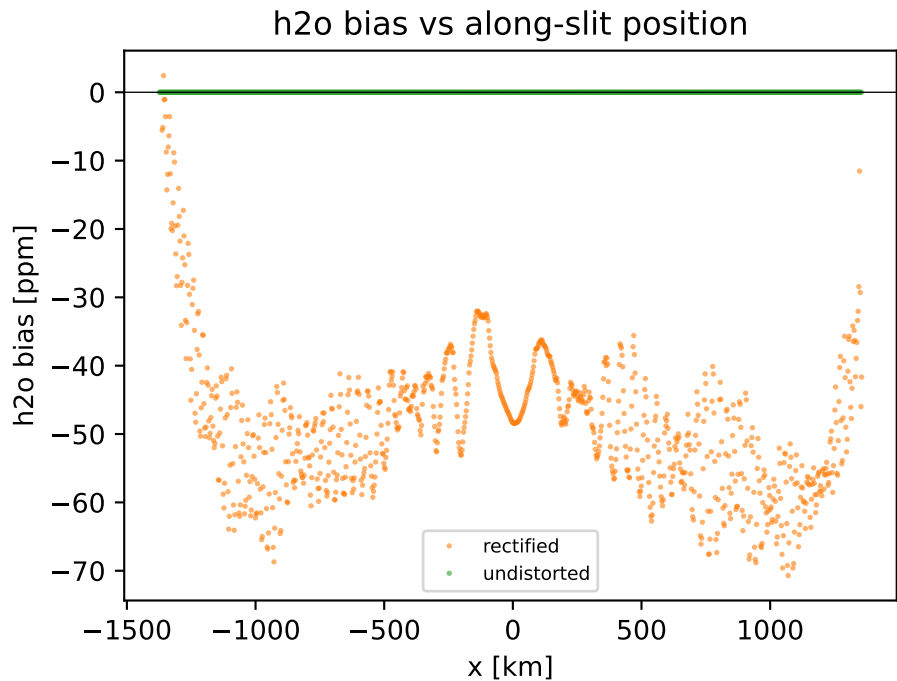
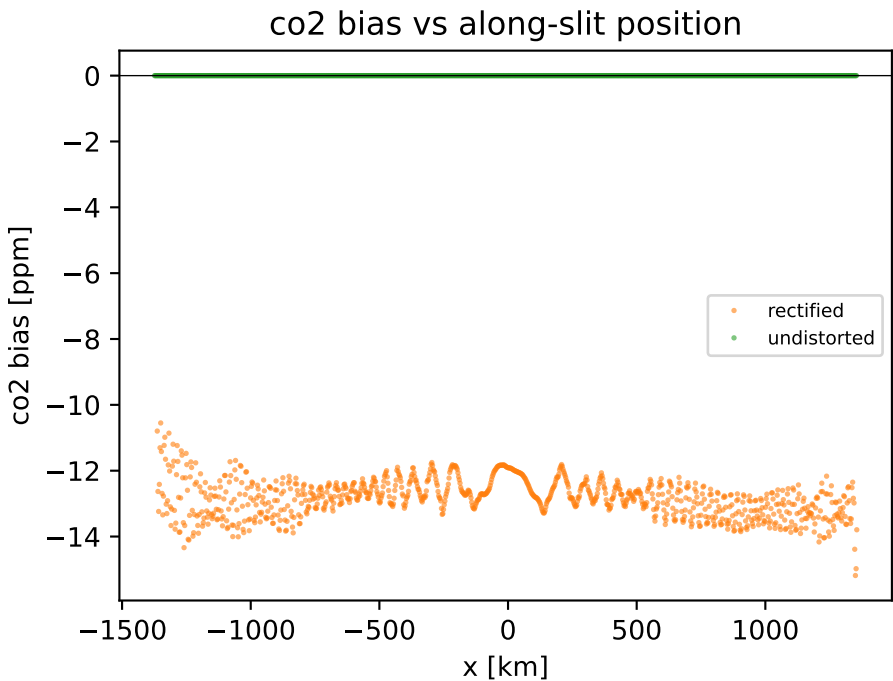


Where each pipeline fails, vs along-slit position

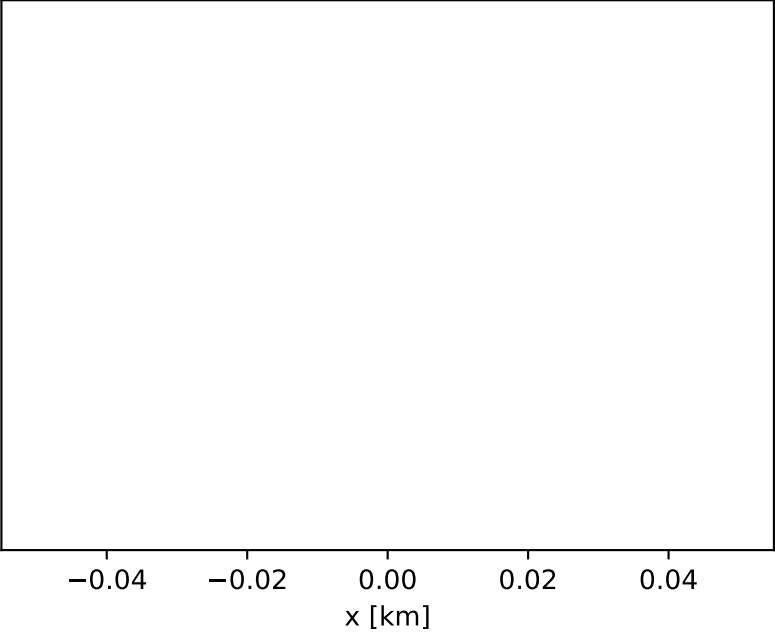


+ rectified stalled (n=6)
rectified off-detector (n=1)

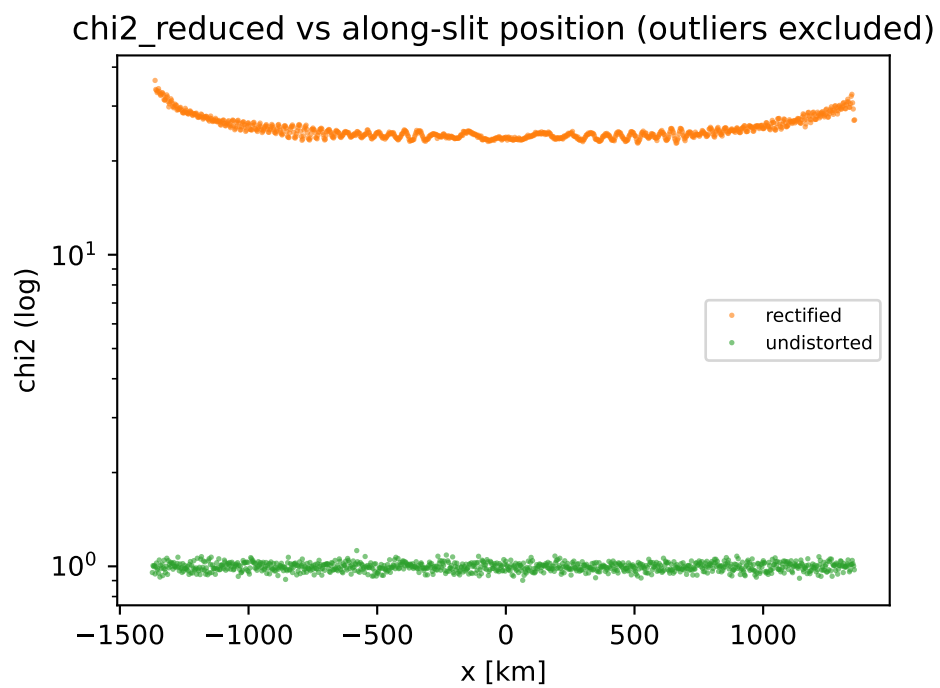
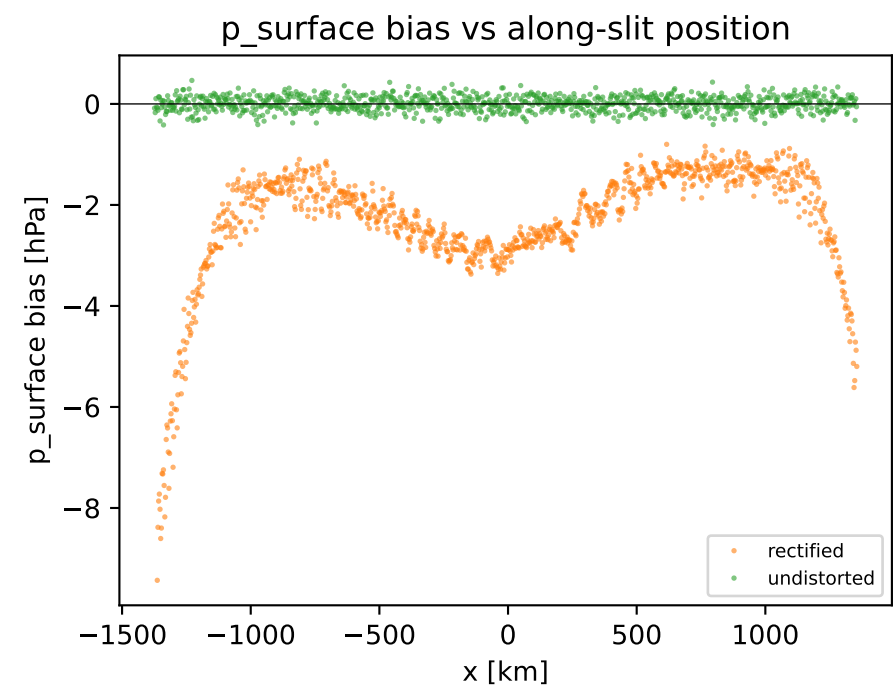
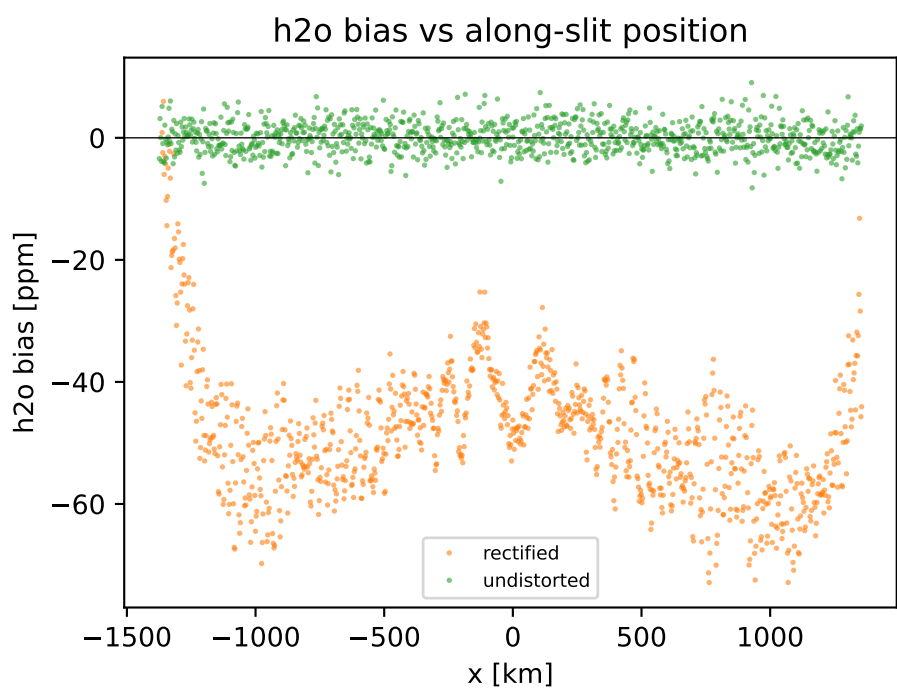
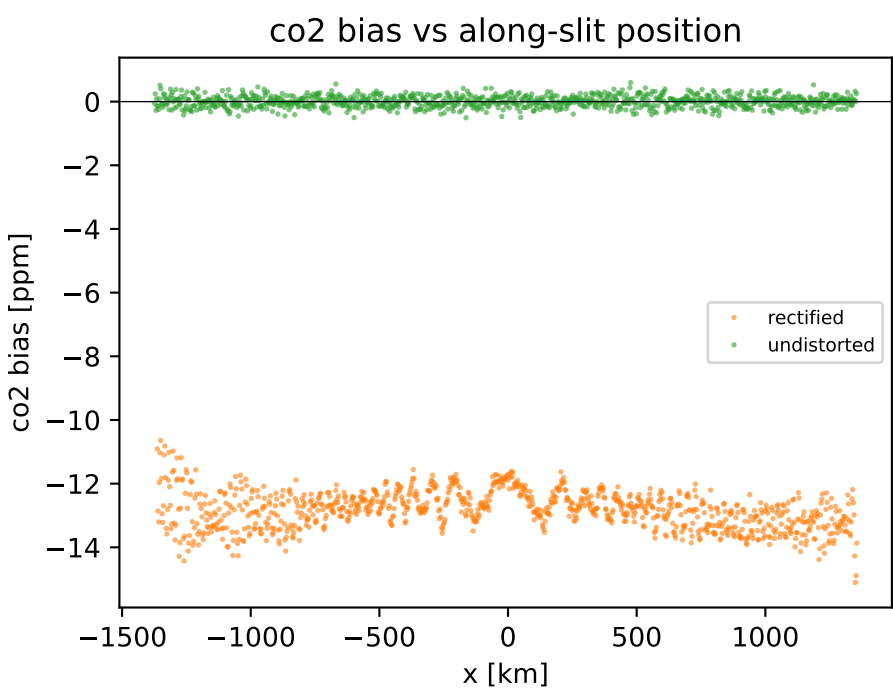
Joint FPA0+FPA2 -- uniform, no noise (rectified/undistorted)



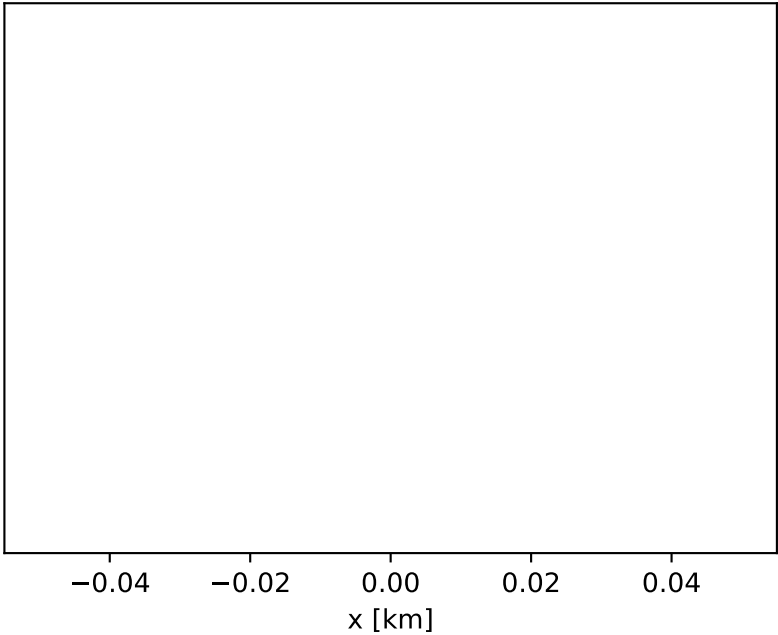
Where each pipeline fails, vs along-slit position



Joint FPA0+FPA2 -- uniform, noise (rectified/undistorted)

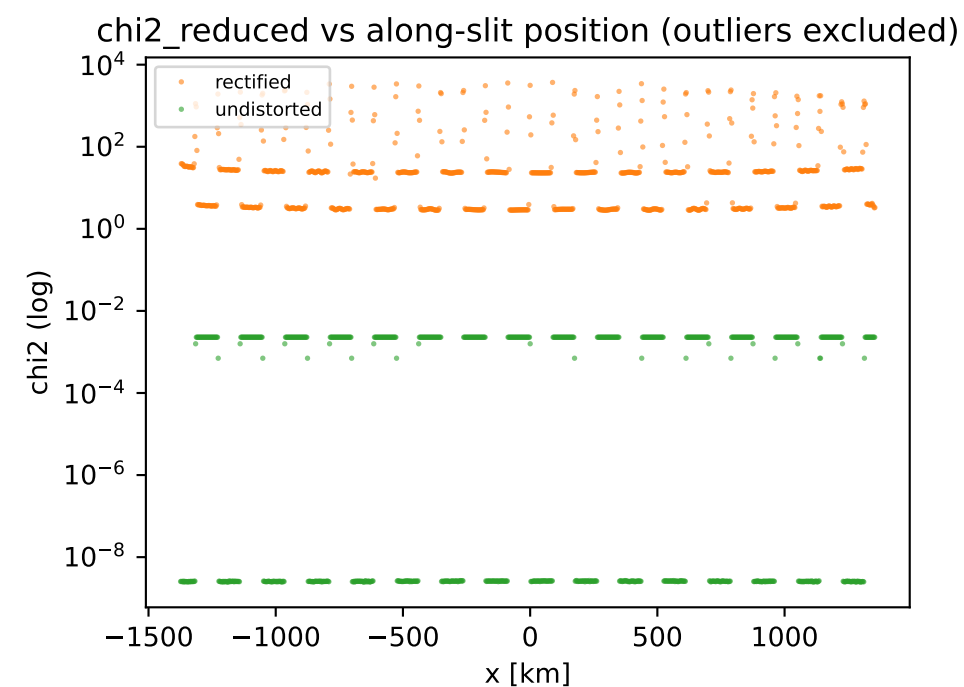
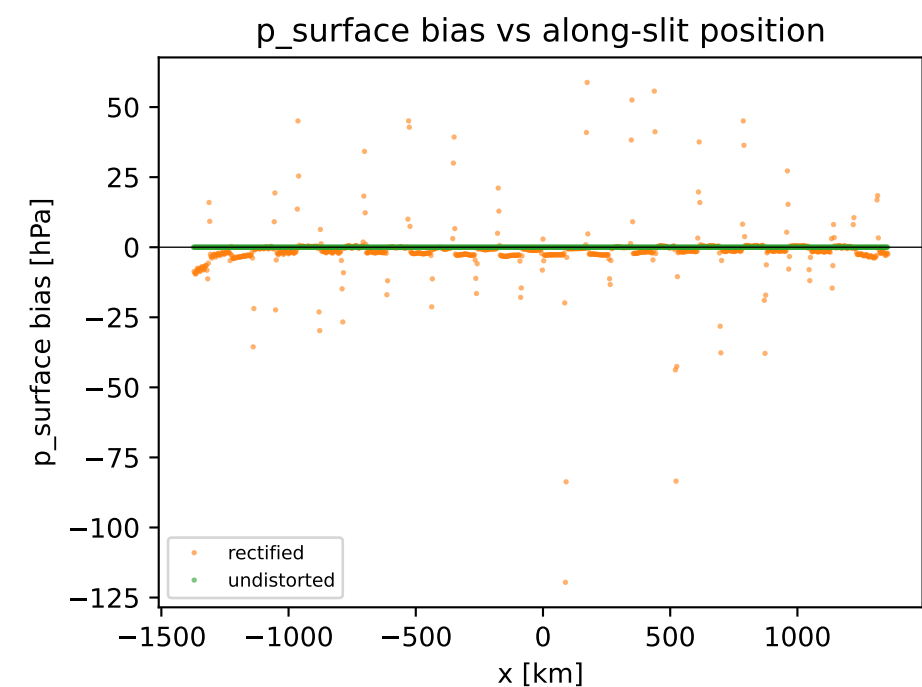
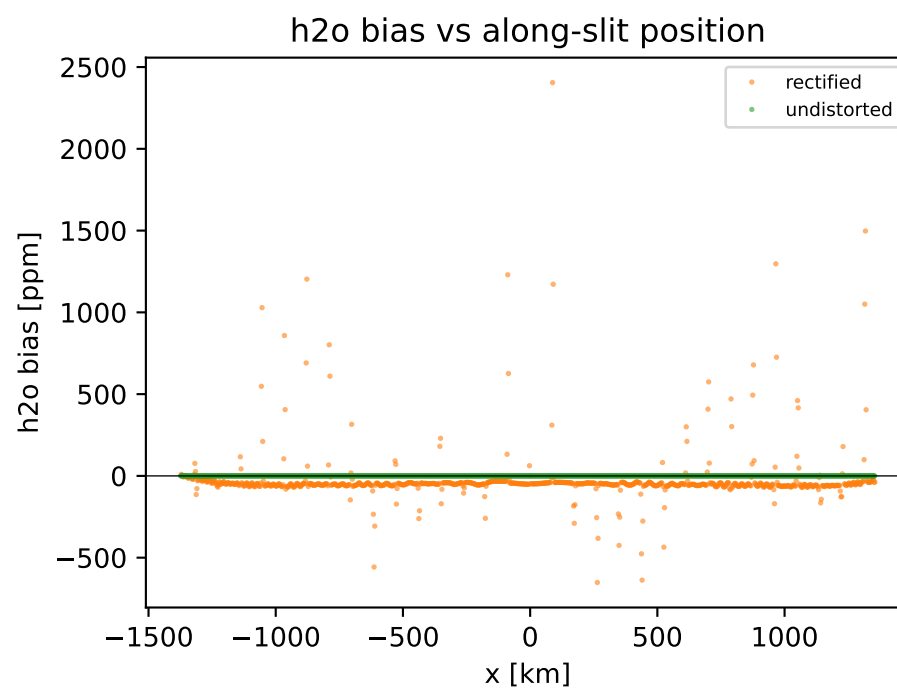
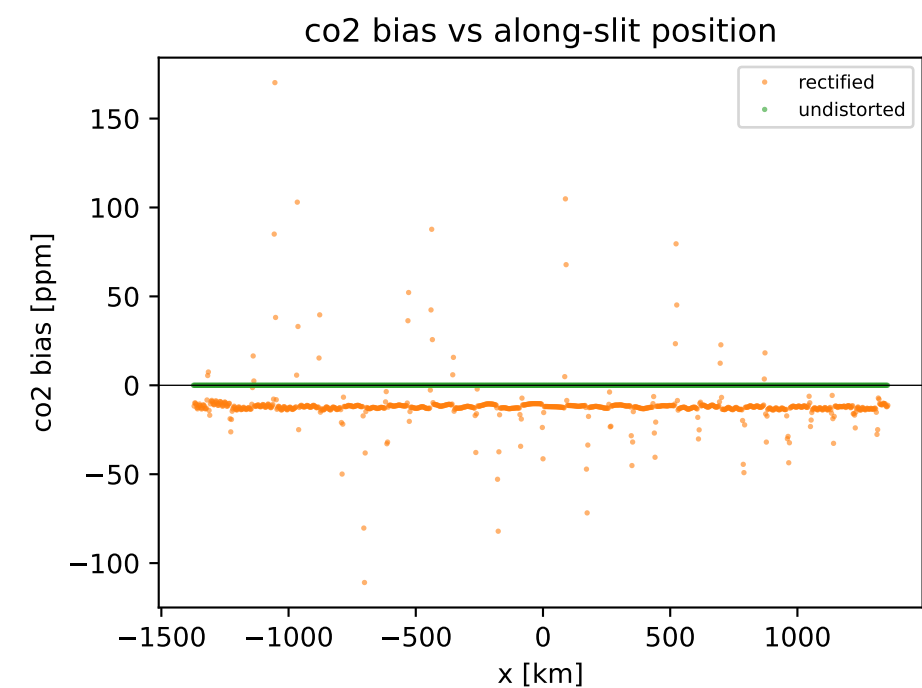


Where each pipeline fails, vs along-slit position

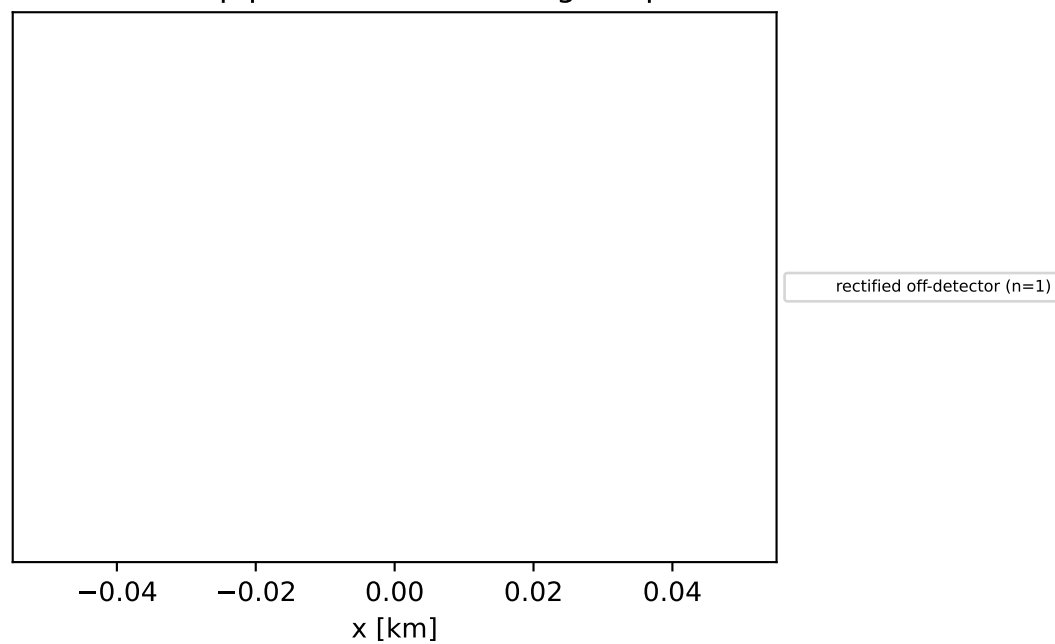


rectified off-detector (n=1)
rectified chi2-outlier (n=3)

Joint FPA0+FPA2 -- barcode, no noise (rectified/undistorted)

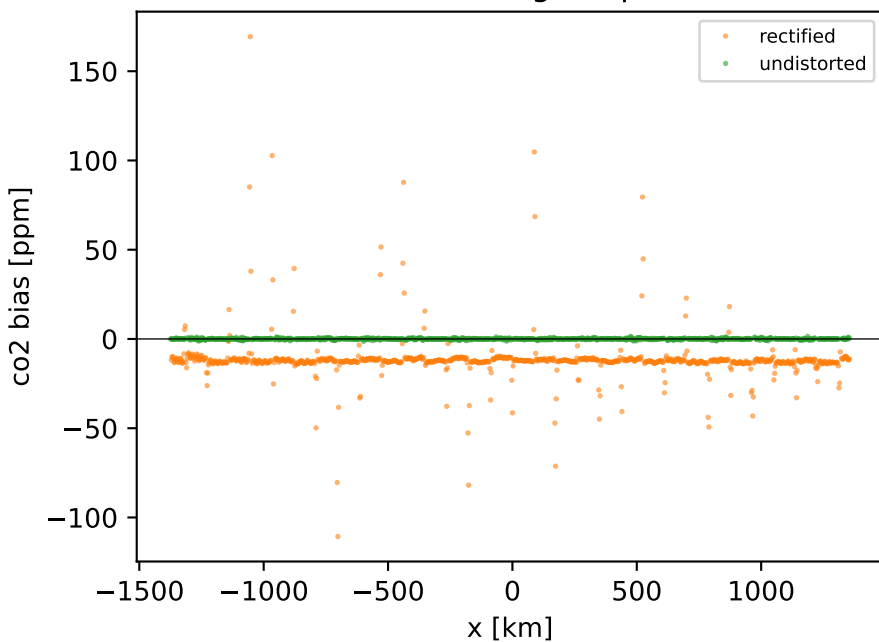


Where each pipeline fails, vs along-slit position

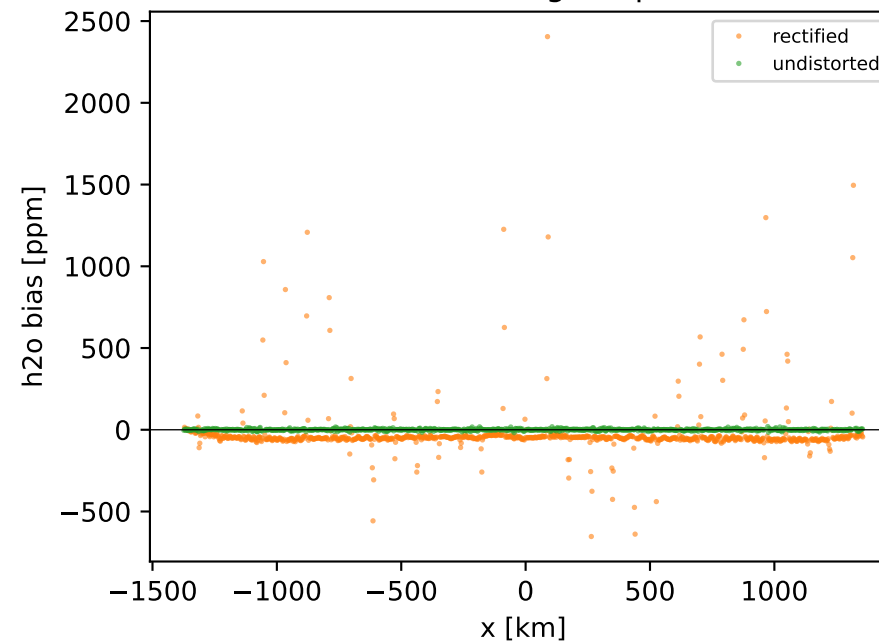


Joint FPA0+FPA2 -- barcode, noise (rectified/undistorted)

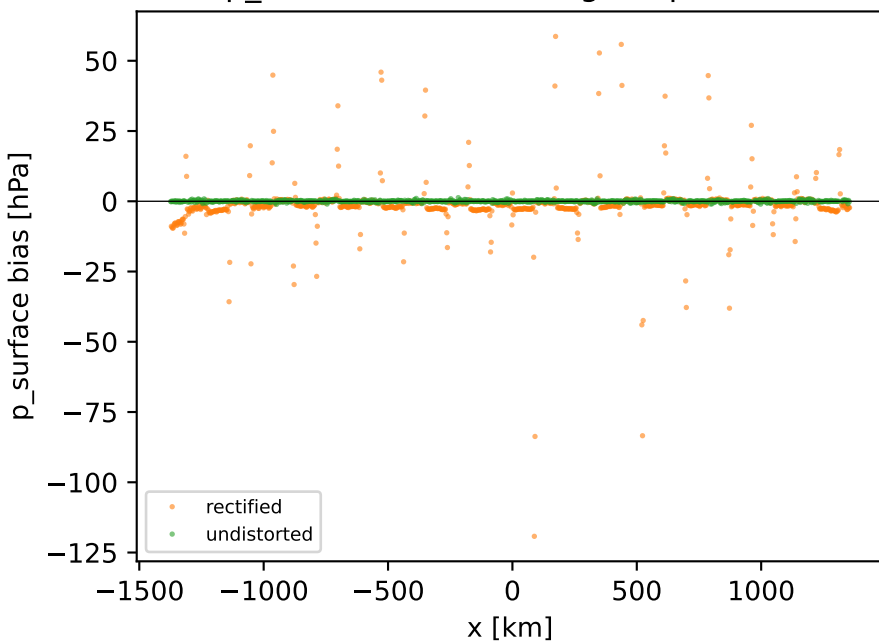
co2 bias vs along-slit position



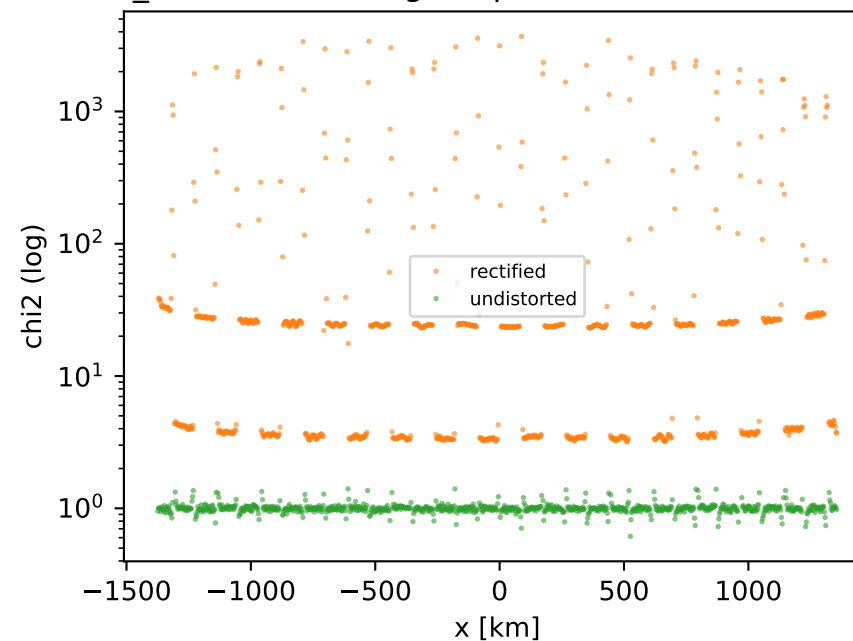
h2o bias vs along-slit position



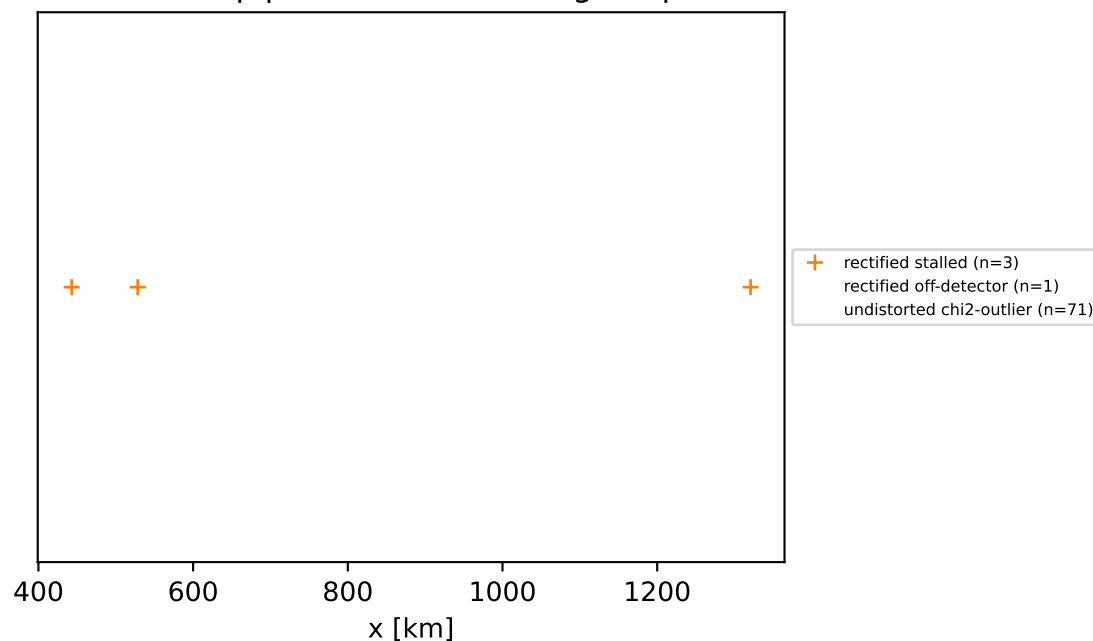
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)



Where each pipeline fails, vs along-slit position



FPA0+FPA2 -- robust bias statistics across scenes/noise (rectified/undistorted)

scene	noise	quantity	pipeline	n	mean	std	median	MAD-sig	n_out
realistic	0	co2	rectified	1017	-12.74	0.9117	-12.65	0.6756	0
realistic	0	h2o	rectified	1017	+334.1	599.2	+253.3	854.6	0
realistic	0	p_surface	rectified	1017	-2.842	6.71	-1.559	1.418	127
realistic	0	co2	undistorted	1007	-0.08471	0.3971	-0.1179	0.02975	208
realistic	0	h2o	undistorted	1007	+371.6	628.2	+274.2	900.4	0
realistic	0	p_surface	undistorted	1007	+0.05551	0.05396	+0.07167	0.06819	0
realistic	1	co2	rectified	1017	-12.74	0.929	-12.64	0.7109	0
realistic	1	h2o	rectified	1017	+334.1	599.1	+255	853.2	0
realistic	1	p_surface	rectified	1017	-2.843	6.714	-1.564	1.438	125
realistic	1	co2	undistorted	1007	-0.08208	0.4319	-0.1182	0.2451	2
realistic	1	h2o	undistorted	1007	+371.6	628.1	+274.5	901	0
realistic	1	p_surface	undistorted	1007	+0.06001	0.1581	+0.06465	0.1698	0
uniform	0	co2	rectified	1020	-12.81	0.5638	-12.8	0.547	0
uniform	0	h2o	rectified	1020	-48.48	10.49	-48.69	9.507	0
uniform	0	p_surface	rectified	1020	-2.32	1.171	-2.087	0.9027	1
uniform	0	co2	undistorted	1007	-0.0001906	3.038e-06	-0.0001906	3.225e-06	0
uniform	0	h2o	undistorted	1007	-0.0004732	3.772e-05	-0.0004747	3.771e-05	0
uniform	0	p_surface	undistorted	1007	-2.937e-05	7.268e-06	-3.113e-05	6.872e-06	0
uniform	1	co2	rectified	1020	-12.81	0.5856	-12.8	0.5912	0
uniform	1	h2o	rectified	1020	-48.42	10.93	-48.88	9.915	0
uniform	1	p_surface	rectified	1020	-2.321	1.178	-2.094	0.8808	1
uniform	1	co2	undistorted	1007	+0.002159	0.1795	+0.004255	0.1839	0
uniform	1	h2o	undistorted	1007	+0.04365	2.48	+0.01964	2.415	0
uniform	1	p_surface	undistorted	1007	+0.004466	0.1458	+0.004795	0.1573	0
barcode	0	co2	rectified	1023	-11.72	12.83	-12.19	1.104	73
barcode	0	h2o	rectified	1023	-29.81	162.2	-49.25	11.13	77
barcode	0	p_surface	rectified	1023	-1.213	8.92	-1.344	1.835	48
barcode	0	co2	undistorted	1007	-0.0001424	5.136e-05	-0.000183	0.0001241	0
barcode	0	h2o	undistorted	1007	-0.0003141	0.0001785	-0.0003819	0.0002391	0
barcode	0	p_surface	undistorted	1007	-4.602e-06	2.701e-05	-7.223e-06	3.768e-05	0
barcode	1	co2	rectified	1020	-11.72	12.83	-12.23	1.208	71
barcode	1	h2o	rectified	1020	-29.62	161.7	-49.22	12.01	72
barcode	1	p_surface	rectified	1020	-1.197	8.94	-1.31	1.806	48
barcode	1	co2	undistorted	936	+0.004272	0.3538	+0.005834	0.2686	0
barcode	1	h2o	undistorted	936	+0.2172	4.738	+0.03426	3.579	0
barcode	1	p_surface	undistorted	936	+0.01634	0.3	+0.009653	0.2197	0